road-crash-school-safety-analysis

September 25, 2026

1 SOUTH AUSTRALIA ROAD CRASH ANALYSIS

1.1 Are Road Crash Patterns Different During School Terms Near Government Schools?

A temporal and spatial analysis of South Australian road crashes, 2020–2024

Location: South Australia
Tools: Python · Pandas · GeoPandas · Matplotlib · SciPy

1.2 PROJECT OVERVIEW

Road crashes in South Australia show clear peaks during morning and afternoon travel periods.

This project investigates whether crash patterns differ between **school-term and non-term weekdays**, whether crashes during key travel periods are more spatially concentrated near **government schools**, and which school areas may warrant further road-safety investigation.

The analysis combines South Australian road crash records from 2020–2024 with government school locations and official South Australian school-term dates.

Rather than simply describing when and where crashes occur, the project examines temporal, spatial, and severity patterns while avoiding causal conclusions that are not supported by the available data.

1.3 ANALYTICAL QUESTIONS

RQ1 - When?

- **How does average road-crash frequency by time of day differ between school-term and non-term weekdays?**
- This analysis compares hourly average crash frequency after accounting for the different numbers of school-term and non-term weekdays.

RQ2 - Where?

- **During key travel periods, are crashes more spatially concentrated near government schools on school-term weekdays than on non-term weekdays?**
- Crash proximity is compared between school-term and non-term weekdays at 8 AM and 3 PM using distance thresholds of 250 m, 500 m, and 1 km from the nearest eligible government school.

RQ3 - Priority

- **Which government school areas show relatively high observed numbers of school-term travel-time crashes and may warrant further investigation?**
- School areas are compared using crash frequency, proximity to the nearest eligible government school, and crash timing during the 8 AM and 3 PM school-travel periods to identify locations that may warrant further investigation.

RQ4 - Severity

- **Are crashes near government schools during school-term travel periods more severe than comparable crashes farther away?**
- Crash severity is compared between crashes occurring within 500 m of the nearest eligible government school and those occurring beyond 500 m.

1.4 SCOPE

- Location: South Australia
- Period: 2020–2024
- Crash data: South Australian road crash records
- School data: South Australian Government education sites
- School calendar: Official South Australian school term dates
- Tools: Python, Pandas, GeoPandas, Matplotlib

1.5 DATA SOURCES

This project uses three data sources:

South Australia Road Crash Data (2020–2024)

- Contains individual crash records including date/time, location, severity, road conditions, crash type, and spatial coordinates.
- Source: Government of South Australia / Data SA
- Temporal coverage: 2020–2024
- Format: CSV
- Accessed: November 2025

South Australian Government Education Sites

- Provides locations and attributes for South Australian Government education sites.
- Source: Government of South Australia / Data SA
- Geographic coverage: South Australia
- Format: GeoJSON
- Accessed: November 2025

For spatial analysis, relevant government school types are retained. Findings therefore relate only to government schools represented in this dataset and should not be interpreted as covering all schools in South Australia.

South Australian School Term Dates (2020–2024)

- Official South Australian school term dates are used to classify crash dates as school-term or non-term periods.
- Source: Government of South Australia
- Temporal coverage: 2020–2024
- Data preparation: Published term dates were manually structured into a pandas DataFrame in the notebook.
- Accessed: November 2025

1.5.1 UNIT OF ANALYSIS

- The primary unit of analysis is an individual road crash (`REPORT_ID`).
- For spatial analysis, each crash is matched to its nearest eligible government school and the distance between the crash and school is calculated in metres.

2 1. SETUP AND DATA LOADING

This section imports the required Python libraries and loads the road crash and school location datasets used throughout the analysis.

2.1 1.1. Import libraries

[0]:
```
import pandas as pd
import geopandas as gpd
import matplotlib.pyplot as plt

from pyproj import CRS
from scipy.stats import chi2_contingency

pd.set_option("display.max_columns", None)
```

[0]:
```
print("Pandas:", pd.__version__)
print("GeoPandas:", gpd.__version__)
```

```
Pandas: 2.2.3
GeoPandas: 1.1.4
```

2.2 1.2. Data loading

[0]:
```
CRASH_PATH = "/Volumes/khanh_project/dataset_project/da_pj_1/
 ↪2020-2024_DATA_SA_Crash.csv"

SCHOOL_PATH = "/Volumes/khanh_project/dataset_project/da_pj_1/
 ↪GovernmentEducationSites_GDA2020.geojson"
```

[0]:
```
crash_raw = pd.read_csv(CRASH_PATH)
```

```
print(f"Crash dataset: {crash_raw.shape[0]:,} rows × {crash_raw.shape[1]}␣
 ↪columns")

display(crash_raw.head())
```

```
Crash dataset: 63,239 rows × 34 columns
```

[0]:
```
schools_raw = gpd.read_file(SCHOOL_PATH)

print(
    f"School dataset: {schools_raw.shape[0]:,} rows × "
    f"{schools_raw.shape[1]} columns"
)

print(f"CRS: {schools_raw.crs}")

display(schools_raw.head())
```

```
School dataset: 1,259 rows × 55 columns
CRS: EPSG:4326
```

	org_num	org_num_text	site_name
0	104	104	Coromandel Valley Primary School
1	106	106	Cowandilla Primary School
2	107	107	Crafers Primary School
3	111	111	Crystal Brook Primary School
4	114	114	Curramulka Primary School

	site_short_name	street_address	suburb	post_code
0	Coromandel Valley PS	339 Main Road	Coromandel Valley	5051
1	Cowandilla PS	21 Jenkins Street	Cowandilla	5033
2	Crafers PS	55 Piccadilly Road	Crafers	5152
3	Crystal Brook PS	7 Mitchell Street	Crystal Brook	5523
4	Curramulka PS	Sixth Street	Curramulka	5580

	site_leader_name	category_name	site_type	site_sub_type
0	Mr Rob Warncken	School	Primary Education	Primary Schools
1	Mrs Katrina Sexton	School	Primary Education	Primary Schools
2	Mr John Nielsen	School	Primary Education	Primary Schools
3	Mr Maciej Jankowski	School	Primary Education	Primary Schools
4	Mrs Robbie Hering	School	Primary Education	Primary Schools

	designation_type_name	association_type_name	partnership
0	Reception to Year 6	Standalone Organisation Unit	Mitcham Hills
1	Reception to Year 6	Standalone Organisation Unit	West Torrens
2	Reception to Year 6	Standalone Organisation Unit	Mount Lofty
3	Reception to Year 6	Standalone Organisation Unit	Upper Mid North
4	Reception to Year 6	Standalone Organisation Unit	Southern Yorke

```
         portfolio    education_director capacity_management_plan  \
0   Mount Barker 2      Ms Tracey Davies                     None
1  Flinders Park 2       Mrs Roslyn Maio                     None
2   Mount Barker 1  Mrs Dannielle Gibson                     None
3            Pirie      Mrs Sally Wright                     None
4           Kadina       Mr David Cowles                     None

  capacity_management_plan_gazzet  fte_enrolments_cy_less1  \
0                            None                   483.0
1                            None                   341.0
2                            None                   221.0
3                            None                   106.0
4                            None                    25.0

   fte_enrolments_cy_less2  fte_enrolments_cy_less3  fte_enrolments_cy_less4  \
0                    489.0                    465.0                    459.0
1                    376.0                    376.0                    400.0
2                    217.0                    209.0                    219.0
3                    112.0                    115.0                    130.0
4                     21.0                     18.0                     23.0

   fte_enrolments_cy_less5  fte_enrolments_cy_less6 electorate_state  \
0                    491.0                    468.0           Waite
1                    430.0                    465.0    West Torrens
2                    272.0                    286.0          Heysen
3                    160.0                    167.0          Stuart
4                     24.0                     24.0        Narungga

  electorate_federal          local_government_area  \
0              Mayo                     Mitcham (C)
1          Adelaide                West Torrens (C)
2              Mayo             Adelaide Hills (DC)
3              Grey  Port Pirie City and Dists (M)
4              Grey            Yorke Peninsula (DC)

                      school_url  index_disadvantage  \
0  http://www.coromandps.sa.edu.au                 7.0
1  http://www.cowandilla.sa.edu.au                 6.0
2   http://www.crafersps.sa.edu.au                 7.0
3  http://www.crystalbps.sa.edu.au                 4.0
4    https://curramulka.sa.edu.au/                 2.0

  preschool_category_ranking intensive_english_language_cent  \
0                       None                               N
1                       None                               Y
2                       None                               N
3                       None                               N
```

```
4                       None                                N

  site_preschools_school_based_fl site_preschools_standalone_flag  \
0                               N                                 N
1                               N                                 N
2                               N                                 N
3                               N                                 N
4                               N                                 N

   associated_school_org_number associated_school_name opened_ind opened_date  \
0                           NaN                  None          Y   01-JAN-22
1                           NaN                  None          Y   01-JAN-22
2                           NaN                  None          Y   01-JAN-22
3                           NaN                  None          Y   01-JAN-22
4                           NaN                  None          Y   01-JAN-22

  closed_date   latitude   longitude  org_num_1  \
0        None -35.033562  138.626758      104.0
1        None -34.934160  138.559877      106.0
2        None -34.993387  138.713480      107.0
3        None -33.352948  138.213543      111.0
4        None -34.696257  137.713037      114.0

                      general_name              association_type  \
0  Coromandel Valley Primary School  Standalone Organisation Unit
1         Cowandilla Primary School  Standalone Organisation Unit
2            Crafers Primary School  Standalone Organisation Unit
3      Crystal Brook Primary School  Standalone Organisation Unit
4         Curramulka Primary School  Standalone Organisation Unit

   workforce_year  fte_teaching_workforce_schools  \
0          2022.0                            27.8
1          2022.0                            30.0
2          2022.0                            14.4
3          2022.0                             9.3
4          2022.0                             3.0

   fte_non_teaching_workforce_scho  icsea_value school_zone ppp_school  \
0                              7.5       1097.0           N          N
1                             13.3       1051.0           N          N
2                              3.2       1109.0           N          N
3                              8.1        977.0           N          N
4                              1.4        935.0           N          N

   stem_funding__m  sustainable_schools_program_fun  \
0              NaN                          25000.0
1              1.0                          25000.0
2              NaN                              0.0
```

```
3              NaN                                0.0
4              NaN                                0.0

   asset_maintenance_program_fundi power_upgrade_project  \
0                          20280.0                     N
1                              0.0                     Y
2                          17485.0                     N
3                              0.0                     N
4                              0.0                     N

  recycled_water_connection                       geometry
0                         N  POINT (138.62677 -35.03355)
1                         N  POINT (138.55988 -34.93415)
2                         N  POINT (138.71349 -34.99337)
3                         N  POINT (138.21355 -33.35293)
4                         N  POINT (137.71304 -34.69624)
```

2.3 1.3. School term dates

Official South Australian school term dates for 2020–2024 are structured below for use in the analysis.

[0]:
```python
school_terms_data = [
    # 2020
    [2020, 1, "2020-01-28", "2020-04-09"],
    [2020, 2, "2020-04-27", "2020-07-03"],
    [2020, 3, "2020-07-20", "2020-09-25"],
    [2020, 4, "2020-10-12", "2020-12-11"],

    # 2021
    [2021, 1, "2021-01-27", "2021-04-09"],
    [2021, 2, "2021-04-27", "2021-07-02"],
    [2021, 3, "2021-07-19", "2021-09-24"],
    [2021, 4, "2021-10-11", "2021-12-10"],

    # 2022
    [2022, 1, "2022-01-31", "2022-04-14"],
    [2022, 2, "2022-05-02", "2022-07-08"],
    [2022, 3, "2022-07-25", "2022-09-30"],
    [2022, 4, "2022-10-17", "2022-12-16"],

    # 2023
    [2023, 1, "2023-01-30", "2023-04-14"],
    [2023, 2, "2023-05-01", "2023-07-07"],
    [2023, 3, "2023-07-24", "2023-09-29"],
    [2023, 4, "2023-10-16", "2023-12-15"],

    # 2024
```

```
    [2024, 1, "2024-01-29", "2024-04-12"],
    [2024, 2, "2024-04-29", "2024-07-05"],
    [2024, 3, "2024-07-22", "2024-09-27"],
    [2024, 4, "2024-10-14", "2024-12-13"],
]

school_terms = pd.DataFrame(
    school_terms_data,
    columns=["Year", "Term", "Start Date", "End Date"]
)

school_terms["Start Date"] = pd.to_datetime(school_terms["Start Date"])
school_terms["End Date"] = pd.to_datetime(school_terms["End Date"])

print(f"School terms: {len(school_terms)} term periods")
display(school_terms)
```

```
School terms: 20 term periods
```

3 2. DATA UNDERSTANDING AND QUALITY ASSESSMENT

This section examines the structure, completeness, uniqueness, and key analytical fields of the source datasets before any cleaning or feature engineering is applied.

3.1 2.1. Data understanding

[0]:
```
print("=== ROAD CRASH DATA ===")
print(f"Rows: {crash_raw.shape[0]:,}")
print(f"Columns: {crash_raw.shape[1]}")
print(f"Unique REPORT_ID: {crash_raw['REPORT_ID'].nunique():,}")
print(f"Duplicate rows: {crash_raw.duplicated().sum():,}")
print()

missing_crash = pd.DataFrame({
    "Column": crash_raw.columns,
    "Missing Count": crash_raw.isna().sum().values,
    "Missing %": (
        crash_raw.isna().sum().values / len(crash_raw) * 100
    ).round(2)
})

missing_crash = (
    missing_crash[missing_crash["Missing Count"] > 0]
    .sort_values("Missing %", ascending=False)
)
```

```
display(missing_crash)
```

```
=== ROAD CRASH DATA ===
Rows: 63,239
Columns: 34
Unique REPORT_ID: 63,239
Duplicate rows: 0
```

[0]:
```
print("=== KEY ANALYTICAL FIELDS ===")

print("\nCrash Date Time:")
print(crash_raw["Crash Date Time"].head())

print("\nSeverity:")
print(crash_raw["CSEF Severity"].value_counts(dropna=False))

print("\nCoordinate completeness:")
print(
    crash_raw[["ACCLOC_X", "ACCLOC_Y"]]
    .isna()
    .sum()
)
```

```
=== KEY ANALYTICAL FIELDS ===

Crash Date Time:
0     1/1/2020 0:49:00
1     1/1/2020 2:00:00
2     1/1/2020 9:40:00
3    1/1/2020 12:30:00
4    1/1/2020 12:41:00
Name: Crash Date Time, dtype: object

Severity:
CSEF Severity
1: PDO      43768
2: MI       15711
3: SI        3323
4: Fatal      437
Name: count, dtype: int64

Coordinate completeness:
ACCLOC_X    9
ACCLOC_Y    9
dtype: int64
```

[0]:
```
print("=== SCHOOL LOCATION DATA ===")
print(f"Rows: {schools_raw.shape[0]:,}")
print(f"Columns: {schools_raw.shape[1]}")
print(f"CRS: {schools_raw.crs}")
print()

print("Geometry types:")
print(schools_raw.geometry.geom_type.value_counts(dropna=False))

print("\nSite types:")
print(schools_raw["site_type"].value_counts(dropna=False))
```

```
=== SCHOOL LOCATION DATA ===
Rows: 1,259
Columns: 55
CRS: EPSG:4326

Geometry types:
Point    1258
None        1
Name: count, dtype: int64

Site types:
site_type
Primary Education             350
Preschools                    332
Occasional Care               104
Support Services               82
Specialist Facilities          80
Primary/Secondary Combined     70
Secondary Education            70
Integrated Centres             62
Family Day Care Schemes        23
Playcentres                    22
Aboriginal/Anangu Schools      17
Rural Care Program             17
Special Education              15
Learning Together @ Home       14
Other                           1
Name: count, dtype: int64
```

3.2 2.2. Quality assessment summary

- Crash IDs are unique and no full-row duplicates were found.
- Crash timestamps are complete and available to minute-level precision.
- Crash severity is fully populated.
- Only 9 crash records lack spatial coordinates.
- Education-site data contains non-school facilities, so relevant school types must be filtered

before spatial analysis.
- The school dataset uses EPSG:4326 and requires projection before distance calculations.

4 3. DATA CLEANING AND PREPARATION

The source datasets are prepared for temporal and spatial analysis. Raw datasets are preserved unchanged, while working copies are cleaned, validated, and restricted to records relevant to the research questions.

4.1 3.1. Data cleaning

[0]:
```
crash = crash_raw.copy()
schools = schools_raw.copy()

crash["Crash Date Time"] = pd.to_datetime(
    crash["Crash Date Time"],
    dayfirst=True,
    errors="coerce"
)

print("Invalid timestamps:", crash["Crash Date Time"].isna().sum())
print("Date range:", crash["Crash Date Time"].min(), "to", crash["Crash Date␣
 ↪Time"].max())
```

```
Invalid timestamps: 0
Date range: 2020-01-01 00:15:00 to 2024-12-31 23:40:00
```

[0]:
```
crash["Crash Date"] = crash["Crash Date Time"].dt.normalize()
crash["Hour"] = crash["Crash Date Time"].dt.hour
crash["Day of Week"] = crash["Crash Date Time"].dt.day_name()
crash["Is Weekday"] = crash["Crash Date Time"].dt.weekday < 5

display(
    crash[
        ["Crash Date Time", "Crash Date", "Hour", "Day of Week", "Is Weekday"]
    ].head()
)
```

[0]:
```
crash["Is School Term"] = False

for _, term in school_terms.iterrows():
    mask = crash["Crash Date"].between(
        term["Start Date"],
        term["End Date"]
    )

    crash.loc[mask, "Is School Term"] = True
```

```
print(crash["Is School Term"].value_counts())
```

```
Is School Term
True     48878
False    14361
Name: count, dtype: int64
```

[0]:
```
calendar = pd.DataFrame({
    "Date": pd.date_range(
        start="2020-01-01",
        end="2024-12-31",
        freq="D"
    )
})

calendar["Is Weekday"] = calendar["Date"].dt.weekday < 5
calendar["Is School Term"] = False

for _, term in school_terms.iterrows():
    mask = calendar["Date"].between(
        term["Start Date"],
        term["End Date"]
    )

    calendar.loc[mask, "Is School Term"] = True

weekday_calendar = calendar[
    calendar["Is Weekday"]
].copy()

print(
    weekday_calendar["Is School Term"]
    .value_counts()
    .rename(index={
        True: "School Term Weekdays",
        False: "Non-Term Weekdays"
    })
)
```

```
Is School Term
School Term Weekdays    994
Non-Term Weekdays       311
Name: count, dtype: int64
```

[0]:
```
school_types = [
    "Primary Education",
```

```
    "Secondary Education",
    "Primary/Secondary Combined",
    "Special Education",
    "Aboriginal/Anangu Schools"
]

analysis_schools = schools[
    schools["site_type"].isin(school_types)
].copy()

print("Education sites before filtering:", len(schools))
print("Schools retained:", len(analysis_schools))

print()
print(analysis_schools["site_type"].value_counts())
```

```
Education sites before filtering: 1259
Schools retained: 522

site_type
Primary Education             350
Primary/Secondary Combined     70
Secondary Education            70
Aboriginal/Anangu Schools      17
Special Education              15
Name: count, dtype: int64
```

[0]:
```
print("Missing geometry:", analysis_schools.geometry.isna().sum())
print("Empty geometry:", analysis_schools.geometry.is_empty.sum())
print("Invalid geometry:", (~analysis_schools.geometry.is_valid).sum())
```

```
Missing geometry: 0
Empty geometry: 0
Invalid geometry: 0
```

[0]:
```
crash_spatial = crash.dropna(
    subset=["ACCLOC_X", "ACCLOC_Y"]
).copy()

print("Crash records:", len(crash))
print("Spatially usable crashes:", len(crash_spatial))
print("Excluded due to missing coordinates:", len(crash) - len(crash_spatial))
```

```
Crash records: 63239
Spatially usable crashes: 63230
Excluded due to missing coordinates: 9
```

[0]:
```
crash_crs = CRS.from_proj4(
    "+proj=lcc "
    "+lat_1=-28 "
    "+lat_2=-36 "
    "+lat_0=-32 "
    "+lon_0=135 "
    "+x_0=1000000 "
    "+y_0=2000000 "
    "+ellps=GRS80 "
    "+units=m "
    "+no_defs "
    "+type=crs"
)

crash_gdf = gpd.GeoDataFrame(
    crash_spatial,
    geometry=gpd.points_from_xy(
        crash_spatial["ACCLOC_X"],
        crash_spatial["ACCLOC_Y"]
    ),
    crs=crash_crs
)

print("Crash points:", len(crash_gdf))
print("Crash CRS:", crash_gdf.crs)
```

```
Crash points: 63230
Crash CRS: +proj=lcc +lat_1=-28 +lat_2=-36 +lat_0=-32 +lon_0=135 +x_0=1000000
+y_0=2000000 +ellps=GRS80 +units=m +no_defs +type=crs
```

[0]:
```
crash_3577 = crash_gdf.to_crs("EPSG:3577")
schools_3577 = analysis_schools.to_crs("EPSG:3577")

print("Crash CRS:", crash_3577.crs)
print("School CRS:", schools_3577.crs)
```

```
Crash CRS: EPSG:3577
School CRS: EPSG:3577
```

4.2 3.2. Nearest eligible government school matching

Each spatially valid crash is matched to its nearest eligible government school. Distance is calculated in metres using the Australian Albers projected coordinate system (EPSG:3577).

Because multiple education records may share the same location, the spatial join can return multiple equally-near matches for a single crash. These ties are collapsed so that the final spatial dataset retains one row per crash.

[0]:
```
nearest_school = gpd.sjoin_nearest(
    crash_3577,
    schools_3577[
        ["site_name", "site_type", "site_sub_type", "geometry"]
    ],
    how="left",
    distance_col="Distance to School"
)

print("Crash points before join:", len(crash_3577))
print("Rows after nearest join:", len(nearest_school))
print("Unique REPORT_ID:", nearest_school["REPORT_ID"].nunique())
```

```
Crash points before join: 63230
Rows after nearest join: 64464
Unique REPORT_ID: 63230
```

[0]:
```
nearest_crash = (
    nearest_school
    .sort_values(["REPORT_ID", "Distance to School"])
    .drop_duplicates(
        subset="REPORT_ID",
        keep="first"
    )
    .copy()
)

print("Final spatial rows:", len(nearest_crash))
print("Unique REPORT_ID:", nearest_crash["REPORT_ID"].nunique())
print(
    "Duplicate REPORT_ID:",
    nearest_crash["REPORT_ID"].duplicated().sum()
)
```

```
Final spatial rows: 63230
Unique REPORT_ID: 63230
Duplicate REPORT_ID: 0
```

[0]:
```
assert len(crash) == 63239
assert crash["REPORT_ID"].is_unique

assert len(analysis_schools) == 522

assert len(nearest_crash) == 63230
assert nearest_crash["REPORT_ID"].is_unique

assert nearest_crash["Distance to School"].notna().all()
```

```
assert (nearest_crash["Distance to School"] >= 0).all()

print("All data preparation checks passed.")
```

```
All data preparation checks passed.
```

4.3 3.3. Data preparation summary

- All 63,239 crash records are retained for temporal analysis.
- 9 records without spatial coordinates are excluded only from spatial analysis.
- 522 relevant government school records are retained from the education-site dataset.
- Crash and school locations are projected to EPSG:3577 for distance calculations.
- Each of the 63,230 spatially valid crashes is matched to its nearest eligible government school.
- Duplicate nearest-school matches caused by tied locations are resolved to preserve one row per crash.
- School-term and weekday indicators are derived for temporal comparisons.

5 4. RQ1 - HOW DO CRASH PATTERNS DIFFER BY TIME OF DAY?

Research Question

How does average road-crash frequency by time of day differ between school-term and non-term weekdays?

To answer this question, crash frequency is first examined across the 24-hour day. School-term and non-term weekdays are then compared using average crashes per eligible weekday rather than raw crash counts, because the dataset contains substantially more school-term weekdays than non-term weekdays.

5.1 4.1. Overall hourly pattern

Road crashes by hour of day - all days, South Australia (2020–2024). The initial hourly overview includes all crash records. Subsequent school-term comparisons are restricted to weekdays.

[0]:
```
hourly_crashes = (
    crash
    .groupby("Hour")
    .size()
    .reindex(range(24), fill_value=0)
)

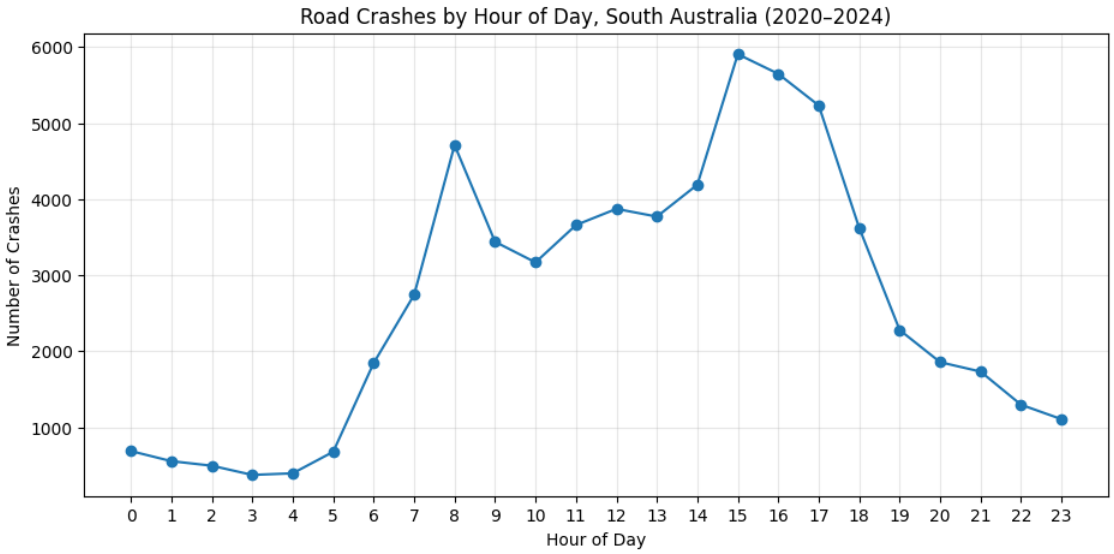
fig, ax = plt.subplots(figsize=(11, 5))

ax.plot(
    hourly_crashes.index,
    hourly_crashes.values,
    marker="o"
```

```
)

ax.set_title(
    "Road Crashes by Hour of Day, South Australia (2020-2024)"
)
ax.set_xlabel("Hour of Day")
ax.set_ylabel("Number of Crashes")
ax.set_xticks(range(24))
ax.grid(alpha=0.3)

plt.show()
```



[0]:
```
display(
    hourly_crashes
    .sort_values(ascending=False)
    .head(5)
    .rename("Crashes")
    .to_frame()
)
```

5.2 4.2. School term vs non-term

[0]:
```
weekday_crashes = crash[
    crash["Is Weekday"]
].copy()

term_days = (
    weekday_calendar["Is School Term"]
```

```
    .value_counts()
    .get(True, 0)
)

nonterm_days = (
    weekday_calendar["Is School Term"]
    .value_counts()
    .get(False, 0)
)

print(f"School-term weekdays: {term_days}")
print(f"Non-term weekdays: {nonterm_days}")
```

```
School-term weekdays: 994
Non-term weekdays: 311
```

5.3 4.3. Key hours

[0]:
```
hourly_term = (
    weekday_crashes[
        weekday_crashes["Is School Term"]
    ]
    .groupby("Hour")
    .size()
    .reindex(range(24), fill_value=0)
)

hourly_nonterm = (
    weekday_crashes[
        ~weekday_crashes["Is School Term"]
    ]
    .groupby("Hour")
    .size()
    .reindex(range(24), fill_value=0)
)

hourly_comparison = pd.DataFrame({
    "Hour": range(24),
    "School Term": hourly_term.values / term_days,
    "Non-Term": hourly_nonterm.values / nonterm_days
})

display(hourly_comparison.round(2))
```

[0]:
```
fig, ax = plt.subplots(figsize=(11, 5))

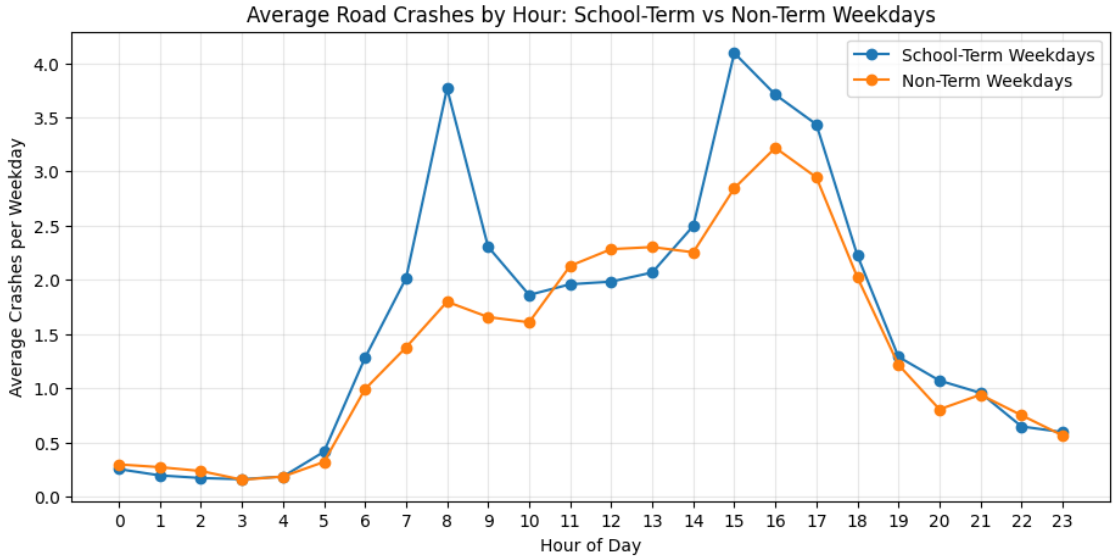
ax.plot(
```

```
    hourly_comparison["Hour"],
    hourly_comparison["School Term"],
    marker="o",
    label="School-Term Weekdays"
)

ax.plot(
    hourly_comparison["Hour"],
    hourly_comparison["Non-Term"],
    marker="o",
    label="Non-Term Weekdays"
)

ax.set_title(
    "Average Road Crashes by Hour: School-Term vs Non-Term Weekdays"
)
ax.set_xlabel("Hour of Day")
ax.set_ylabel("Average Crashes per Weekday")
ax.set_xticks(range(24))
ax.grid(alpha=0.3)
ax.legend()

plt.show()
```



[0]:
```
focus_hours = hourly_comparison[
    hourly_comparison["Hour"].isin([8, 15])
].copy()
```

```
focus_hours["Relative Difference %"] = (
    (
        focus_hours["School Term"]
        / focus_hours["Non-Term"]
        - 1
    ) * 100
)

focus_hours["Frequency Ratio"] = (
    focus_hours["School Term"]
    / focus_hours["Non-Term"]
)

display(
    focus_hours[
        [
            "Hour",
            "School Term",
            "Non-Term",
            "Frequency Ratio",
            "Relative Difference %"
        ]
    ].round(2)
)
```

[0]:
```
hourly_comparison["Frequency Ratio"] = (
    hourly_comparison["School Term"]
    / hourly_comparison["Non-Term"]
)

hourly_comparison["Absolute Difference"] = (
    hourly_comparison["School Term"]
    - hourly_comparison["Non-Term"]
)
```

5.4 4.4. RQ1 - Key finding

- Crash frequency shows a pronounced school-term elevation at 8 AM. School-term weekdays record an average of **3.77 crashes at 8 AM per day**, compared with **1.80 on non-term weekdays**, representing an **approximately 2.1 times the non-term level (+109.6%)**.
- This difference is substantially larger than at the surrounding morning hours. In contrast, school-term crash frequency is similar to or lower than non-term frequency around midday, indicating that the school-term difference is not simply a uniform increase throughout the day.
- A second difference appears at 3 PM, where school-term weekdays average **4.09 crashes per day** compared with **2.85 during non-term weekdays** (**1.44×; +43.9%**). However,

afternoon crash activity is elevated in both groups, so this pattern may also reflect broader afternoon traffic activity.

- Overall, the temporal evidence is strongest at **8 AM**, while the 3 PM pattern warrants further spatial investigation before being interpreted as school-related.

5.5 4.5. Year-by-year robustness check

Because the analysis covers 2020–2024, the school-term effect is also examined by year to assess whether the overall pattern is persistent or driven primarily by a particular year.

[0]:
```
rq1_year = crash[
    crash["Is Weekday"] &
    crash["Hour"].isin([8, 15])
].copy()

rq1_year["Period"] = rq1_year["Hour"].map({
    8: "8 AM",
    15: "3 PM"
})

rq1_year["Year"] = rq1_year["Crash Date"].dt.year

yearly_crashes = (
    rq1_year
    .groupby(["Year", "Period", "Is School Term"])
    .size()
    .reset_index(name="Crashes")
)

display(yearly_crashes)
```

[0]:
```
calendar["Year"] = calendar["Date"].dt.year

yearly_days = (
    calendar[calendar["Is Weekday"]]
    .groupby(["Year", "Is School Term"])
    .size()
    .reset_index(name="Eligible Weekdays")
)

yearly_rates = yearly_crashes.merge(
    yearly_days,
    on=["Year", "Is School Term"],
    how="left"
)

yearly_rates["Avg Crashes per Weekday"] = (
```

```
    yearly_rates["Crashes"] /
    yearly_rates["Eligible Weekdays"]
)

yearly_rates["Group"] = yearly_rates["Is School Term"].map({
    True: "School Term",
    False: "Non-Term"
})

display(
    yearly_rates[
        [
            "Year",
            "Period",
            "Group",
            "Crashes",
            "Eligible Weekdays",
            "Avg Crashes per Weekday"
        ]
    ].round(2)
)
```

- The school-term pattern is consistent across all five years from 2020 to 2024.
- At **8 AM**, average crashes per weekday were higher during school terms than during non-term periods in every year. The same direction was observed at **3 PM** across all five years.
- This indicates that the overall temporal findings are not driven solely by a single year and remain visible across the full 2020–2024 study period.
- The magnitude of the difference varies from year to year, so the pooled 2020–2024 results should be interpreted as an overall pattern rather than a constant effect size.

6 5. RQ2 - ARE SCHOOL-TRAVEL-TIME CRASHES MORE CONCENTRATED NEAR SCHOOLS?

Research Question

Are crashes during school travel periods more concentrated near government schools during school terms than during non-term periods?

The temporal analysis identified a particularly strong school-term signal at 8 AM and a smaller signal at 3 PM. This section examines whether those temporal differences are also associated with crash proximity to the nearest eligible government school.

Descriptive comparisons use cumulative distance thresholds of 250 m, 500 m, and 1 km. For statistical testing, crash distances are divided into mutually exclusive bands (0–250 m, 250–500 m, 500–1000 m, and >1000 m) to compare the overall spatial distribution between school-term and non-term crashes.

6.1 5.1. Distance overview

[0]:
```
distance_summary = (
    nearest_crash["Distance to School"]
    .describe(
        percentiles=[0.25, 0.5, 0.75, 0.9, 0.95]
    )
)

display(distance_summary.to_frame("Distance (m)").round(1))
```

[0]:
```
distance_thresholds = []

for distance in [250, 500, 1000, 2000]:
    count = (
        nearest_crash["Distance to School"] <= distance
    ).sum()

    distance_thresholds.append({
        "Distance": f" {distance:,} m",
        "Crashes": count,
        "Percentage": count / len(nearest_crash) * 100
    })

distance_threshold_summary = pd.DataFrame(distance_thresholds)

display(distance_threshold_summary.round(2))
```

6.2 5.2. Term and non-term comparison

[0]:
```
rq2_data = nearest_crash[
    (nearest_crash["Is Weekday"]) &
    (nearest_crash["Hour"].isin([8, 15]))
].copy()

rq2_data["Period"] = rq2_data["Hour"].map({
    8: "8 AM",
    15: "3 PM"
})
```

[0]:
```
rq2_results = []

for period in ["8 AM", "3 PM"]:
    for is_term in [True, False]:

        subset = rq2_data[
            (rq2_data["Period"] == period) &
```

```
            (rq2_data["Is School Term"] == is_term)
        ]

        row = {
            "Period": period,
            "Group": "School Term" if is_term else "Non-Term",
            "Total Crashes": len(subset)
        }

        for distance in [250, 500, 1000]:
            row[f"Within {distance}m %"] = (
                (subset["Distance to School"] <= distance)
                .mean() * 100
            )

        rq2_results.append(row)

rq2_summary = pd.DataFrame(rq2_results)

display(rq2_summary.round(2))
```

6.3 5.3. Statistical test

Because 8 AM and 3 PM were identified from the temporal analysis in the same dataset, these statistical tests are interpreted as exploratory evidence.

[0]:
```
# Create mutually exclusive distance bands
rq2_data["Distance Band"] = pd.cut(
    rq2_data["Distance to School"],
    bins=[0, 250, 500, 1000, float("inf")],
    labels=[
        "0-250 m",
        "250-500 m",
        "500-1000 m",
        ">1000 m"
    ],
    include_lowest=True
)

stat_results = []

for period in ["8 AM", "3 PM"]:

    period_data = rq2_data[
        rq2_data["Period"] == period
    ]
```

```
    table = pd.crosstab(
        period_data["Is School Term"],
        period_data["Distance Band"]
    )

    chi2, p_value, dof, expected = chi2_contingency(table)

    stat_results.append({
        "Period": period,
        "Chi-square": chi2,
        "Degrees of Freedom": dof,
        "p-value": p_value
    })

stat_test_results = pd.DataFrame(stat_results)

display(stat_test_results.round(4))
```

[0]:
```
effect_results = []

for period in ["8 AM", "3 PM"]:

    period_rows = rq2_summary[
        rq2_summary["Period"] == period
    ]

    term = period_rows[
        period_rows["Group"] == "School Term"
    ].iloc[0]

    nonterm = period_rows[
        period_rows["Group"] == "Non-Term"
    ].iloc[0]

    for distance in [250, 500, 1000]:

        col = f"Within {distance}m %"

        effect_results.append({
            "Period": period,
            "Distance": f" {distance} m",
            "School Term %": term[col],
            "Non-Term %": nonterm[col],
            "Difference (pp)": term[col] - nonterm[col]
        })

effect_summary = pd.DataFrame(effect_results)
```

```
display(effect_summary.round(2))
```

6.4 5.4. Visualization - Spatial Comparison

The following comparison examines the proportion of crashes occurring within 250 m, 500 m, and 1 km of the nearest eligible government school during the two school-travel periods identified in the temporal analysis.

[0]:
```
plot_data = effect_summary.copy()

fig, axes = plt.subplots(1, 2, figsize=(12, 5), sharey=True)

for ax, period in zip(axes, ["8 AM", "3 PM"]):

    subset = plot_data[
        plot_data["Period"] == period
    ]

    x = range(len(subset))
    width = 0.35

    ax.bar(
        [i - width/2 for i in x],
        subset["School Term %"],
        width,
        label="School Term"
    )

    ax.bar(
        [i + width/2 for i in x],
        subset["Non-Term %"],
        width,
        label="Non-Term"
    )

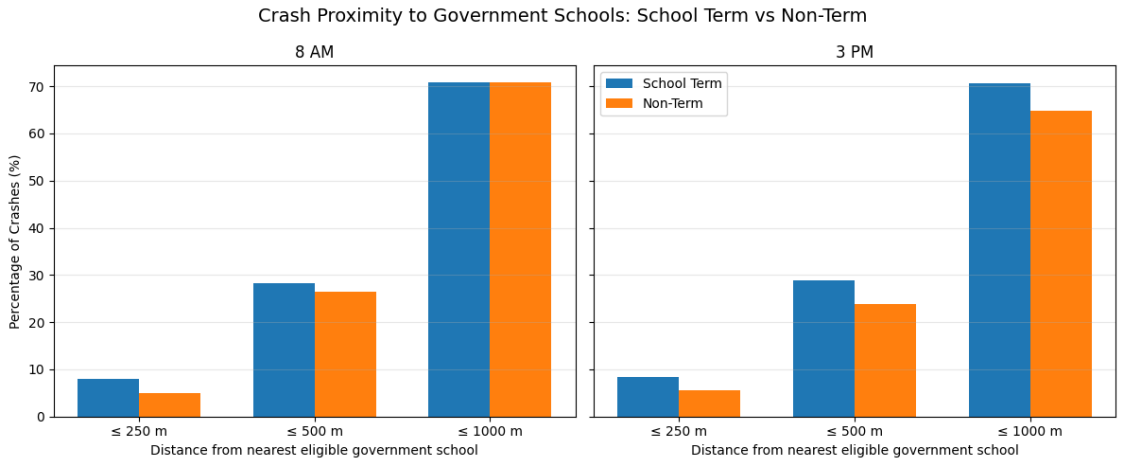
    ax.set_xticks(list(x))
    ax.set_xticklabels(subset["Distance"])
    ax.set_title(period)
    ax.set_xlabel("Distance from nearest eligible government school")
    ax.grid(axis="y", alpha=0.3)

axes[0].set_ylabel("Percentage of Crashes (%)")
axes[1].legend()

fig.suptitle(
    "Crash Proximity to Government Schools: School Term vs Non-Term",
```

```
    fontsize=14
)

plt.tight_layout()
plt.show()
```



Morning and afternoon school-travel periods show different patterns. The 8 AM period shows a stronger temporal difference between school-term and non-term weekdays, while the 3 PM period shows a clearer spatial proximity pattern around government schools.

This suggests that the school-term crash pattern is not uniform across the day and that morning and afternoon travel periods may warrant different forms of further investigation.

6.5 5.5. RQ2 - Key finding

- The spatial analysis reveals different patterns during the morning and afternoon school-travel periods.
- At **8 AM**, school-term crashes were more concentrated within **250 m** of the nearest eligible government school (8.07%) than non-term crashes (5.01%), a difference of **3.06 percentage points**. However, when crash distances were divided into mutually exclusive distance bands (0–250 m, 250–500 m, 500–1000 m, and >1000 m), the overall difference in the distance distribution was **not statistically significant** ($^2 = 6.68$, df = 3, p = 0.0828).
- At **3 PM**, school-term crashes showed greater proximity to schools: 8.41% occurred within 250 m compared with 5.54% during non-term periods, 28.83% vs 23.84% within 500 m, and 70.62% vs 64.86% within 1 km. The overall distance distribution differed **significantly** between school-term and non-term crashes ($^2 = 17.27$, df = 3, p = 0.0006).
- Combined with the temporal results, the findings suggest two distinct patterns: **8 AM shows the strongest increase in overall crash frequency during school terms, whereas 3 PM provides stronger statistical evidence that crashes are more spatially concentrated around schools.**

- These results indicate an association between school terms, school-travel periods, and crash patterns, but they do not establish that school activity directly causes the crashes.

7 6. RQ3 - WHICH GOVERNMENT SCHOOL AREAS MAY WARRANT FURTHER INVESTIGATION?

Research Question

Which government school areas have relatively high observed numbers of school-term travel-time crashes within 500 m and may warrant further investigation?

This analysis focuses on weekday crashes occurring at 8 AM and 3 PM during school terms. A 500 m radius from the nearest eligible government school is used as the primary hotspot threshold, while the 250 m threshold provides additional context for crashes occurring very close to school sites.

School-level counts represent crashes geographically assigned to the nearest eligible government school and are used to identify priority areas for further investigation rather than as measures of individual school safety.

7.1 6.1. Unique physical school locations

[0]:
```
# Create unique school locations for hotspot analysis.
# Multiple education records may share the same physical location.

school_locations = (
    schools_3577
    .groupby("geometry")
    .agg(
        site_name=(
            "site_name",
            lambda x: " / ".join(sorted(set(x.dropna())))
        ),
        site_type=(
            "site_type",
            lambda x: " / ".join(sorted(set(x.dropna())))
        )
    )
    .reset_index()
)

school_locations = gpd.GeoDataFrame(
    school_locations,
    geometry="geometry",
    crs=schools_3577.crs
)

print("Original school records:", len(schools_3577))
```

```
print("Unique school locations:", len(school_locations))
```

```
Original school records: 522
Unique school locations: 513
```

7.2 6.2. Nearest-location matching

[0]:
```
rq3_nearest = gpd.sjoin_nearest(
    crash_3577,
    school_locations[
        ["site_name", "site_type", "geometry"]
    ],
    how="left",
    distance_col="Distance to School"
)

rq3_nearest = (
    rq3_nearest
    .sort_values(["REPORT_ID", "Distance to School"])
    .drop_duplicates(
        subset="REPORT_ID",
        keep="first"
    )
    .copy()
)

print("RQ3 spatial rows:", len(rq3_nearest))
print("Unique crashes:", rq3_nearest["REPORT_ID"].nunique())
print(
    "Duplicate crash IDs:",
    rq3_nearest["REPORT_ID"].duplicated().sum()
)
```

```
RQ3 spatial rows: 63230
Unique crashes: 63230
Duplicate crash IDs: 0
```

[0]:
```
rq3_data = rq3_nearest[
    (rq3_nearest["Is Weekday"]) &
    (rq3_nearest["Is School Term"]) &
    (rq3_nearest["Hour"].isin([8, 15])) &
    (rq3_nearest["Distance to School"] <= 500)
].copy()

rq3_data["Period"] = rq3_data["Hour"].map({
    8: "8 AM",
    15: "3 PM"
```

```
})

print("Crashes included:", len(rq3_data))
print(
    "Unique school locations:",
    rq3_data["site_name"].nunique()
)
```

```
Crashes included: 2233
Unique school locations: 332
```

7.3 6.3. Hotspot summary

[0]:
```
school_hotspots = (
    rq3_data
    .groupby(
        ["site_name", "site_type"],
        as_index=False
    )
    .agg(
        Total_Crashes=("REPORT_ID", "count"),
        Crashes_8AM=("Hour", lambda x: (x == 8).sum()),
        Crashes_3PM=("Hour", lambda x: (x == 15).sum()),
        Median_Distance_m=("Distance to School", "median")
    )
    .sort_values(
        "Total_Crashes",
        ascending=False
    )
)

display(
    school_hotspots
    .head(15)
    .round(1)
)
```

[0]:
```
very_near_school = (
    rq3_data[
        rq3_data["Distance to School"] <= 250
    ]
    .groupby("site_name")
    .size()
    .rename("Within_250m")
)

school_hotspots = school_hotspots.merge(
```

```
    very_near_school,
    on="site_name",
    how="left"
)

school_hotspots["Within_250m"] = (
    school_hotspots["Within_250m"]
    .fillna(0)
    .astype(int)
)

display(
    school_hotspots[
        [
            "site_name",
            "site_type",
            "Total_Crashes",
            "Within_250m",
            "Crashes_8AM",
            "Crashes_3PM",
            "Median_Distance_m"
        ]
    ]
    .head(15)
    .round(1)
)
```

[0]:
```
school_hotspots["Within_250m_Pct"] = (
    school_hotspots["Within_250m"]
    / school_hotspots["Total_Crashes"]
    * 100
)

top_hotspots = (
    school_hotspots
    .head(15)
    .copy()
)

display(
    top_hotspots[
        [
            "site_name",
            "Total_Crashes",
            "Within_250m",
            "Within_250m_Pct",
            "Crashes_8AM",
```

```
            "Crashes_3PM",
            "Median_Distance_m"
        ]
    ].round(1)
)
```

7.4 6.4. Top 10 visualization

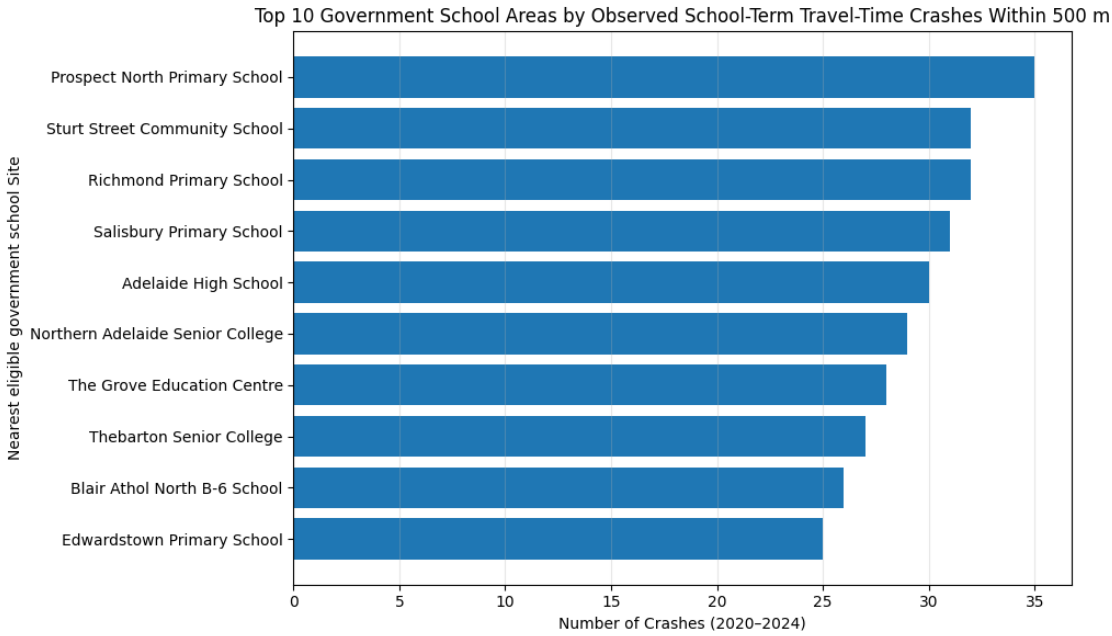
[0]:
```
top10 = (
    school_hotspots
    .head(10)
    .sort_values("Total_Crashes")
)

fig, ax = plt.subplots(figsize=(10, 6))

ax.barh(
    top10["site_name"],
    top10["Total_Crashes"]
)

ax.set_title(
    "Top 10 Government School Areas by Observed School-Term Travel-Time Crashes␣
 ↪Within 500 m")
ax.set_xlabel("Number of Crashes (2020-2024)")
ax.set_ylabel("Nearest eligible government school Site")
ax.grid(axis="x", alpha=0.3)

plt.tight_layout()
plt.show()
```



7.5 6.5. RQ3 - Key finding

- The hotspot analysis identified several school areas with relatively high numbers of school-term weekday crashes occurring at 8 AM or 3 PM within 500 m of the nearest eligible government school.
- **Prospect North Primary School** recorded the highest observed count, with 35 crashes, followed by **Sturt Street Community School** and **Richmond Primary School**, with 32 crashes each.
- However, total crash count alone does not describe how tightly crashes are concentrated around a school. For example, **Northern Adelaide Senior College** recorded 29 crashes within 500 m, of which 19 occurred within 250 m, with a median distance of approximately 187 m. In comparison, Prospect North Primary School recorded more crashes overall but only 6 occurred within 250 m and its median distance was approximately 370 m.
- The timing of crashes also differs between school areas. Some locations show a stronger morning pattern, while others have more crashes during the 3 PM period.
- These rankings should therefore be interpreted as **priority areas for further investigation rather than rankings of school safety**. Crash counts may also reflect surrounding road networks, traffic volumes, land use, and other environmental factors that are not controlled for in this analysis.

8 7. RQ4 - ARE CRASHES NEAR GOVERNMENT SCHOOLS MORE SEVERE?

Research Question

During school-term travel periods, are crashes occurring near schools more severe than crashes occurring farther away?

To make the comparison more meaningful, the analysis is restricted to school-term weekdays at 8 AM and 3 PM. Crashes within 500 m of the nearest eligible government school are compared with crashes occurring more than 500 m away.

8.1 7.1. Severity distribution

[0]:
```
rq4_data = nearest_crash[
    (nearest_crash["Is Weekday"]) &
    (nearest_crash["Is School Term"]) &
    (nearest_crash["Hour"].isin([8, 15]))
].copy()

rq4_data["Location Group"] = (
    rq4_data["Distance to School"]
    .le(500)
    .map({
        True: "Within 500 m",
        False: "Beyond 500 m"
    })
)

print("RQ4 crashes:", len(rq4_data))

display(
    rq4_data["Location Group"]
    .value_counts()
    .to_frame("Crashes")
)
```

```
RQ4 crashes: 7812
```

[0]:
```
severity_table = pd.crosstab(
    rq4_data["Location Group"],
    rq4_data["CSEF Severity"]
)

severity_order = [
    "1: PDO",
    "2: MI",
    "3: SI",
    "4: Fatal"
]

severity_table = severity_table.reindex(
    columns=severity_order,
```

```
    fill_value=0
)

display(severity_table)
```

[0]:
```
severity_pct = (
    severity_table
    .div(severity_table.sum(axis=1), axis=0)
    * 100
)

display(severity_pct.round(2))
```

8.2 7.2. Injury/Fatal comparison

[0]:
```
rq4_data["Injury_or_Fatal"] = (
    rq4_data["CSEF Severity"] != "1: PDO"
)

injury_summary = (
    rq4_data
    .groupby("Location Group")
    .agg(
        Total_Crashes=("REPORT_ID", "count"),
        Injury_or_Fatal_Crashes=("Injury_or_Fatal", "sum")
    )
)

injury_summary["Injury_or_Fatal_%"] = (
    injury_summary["Injury_or_Fatal_Crashes"]
    / injury_summary["Total_Crashes"]
    * 100
)

display(injury_summary.round(2))
```

8.3 7.3. Statistical test

[0]:
```
severity_binary_table = pd.crosstab(
    rq4_data["Location Group"],
    rq4_data["Injury_or_Fatal"]
)

display(severity_binary_table)

chi2, p_value, dof, expected = chi2_contingency(
```

```
    severity_binary_table
)

print(f"Chi-square: {chi2:.4f}")
print(f"p-value: {p_value:.4f}")
```

```
Chi-square: 0.0017
p-value: 0.9671
```

8.4 7.4. Visualization

[0]:
```
injury_plot = (
    injury_summary
    .reset_index()
)

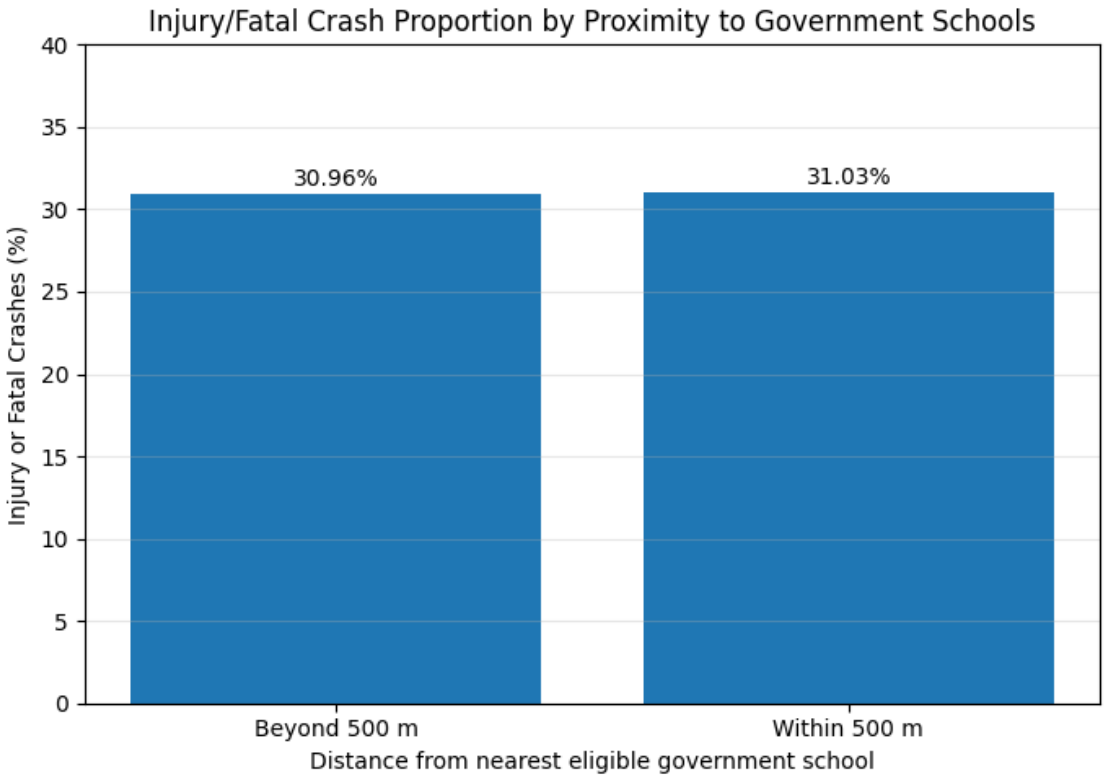
fig, ax = plt.subplots(figsize=(7, 5))

bars = ax.bar(
    injury_plot["Location Group"],
    injury_plot["Injury_or_Fatal_%"]
)

ax.set_title(
    "Injury/Fatal Crash Proportion by Proximity to Government Schools")
ax.set_xlabel("Distance from nearest eligible government school")
ax.set_ylabel("Injury or Fatal Crashes (%)")
ax.set_ylim(0, 40)
ax.grid(axis="y", alpha=0.3)

for bar, value in zip(
    bars,
    injury_plot["Injury_or_Fatal_%"]
):
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 0.5,
        f"{value:.2f}%",
        ha="center"
    )

plt.tight_layout()
plt.show()
```



8.5 7.5. RQ4 - Key finding

- Crash severity was nearly identical between crashes occurring within 500 m of a school and those occurring farther away during school-term weekday travel periods.
- Approximately **31.03%** of crashes within 500 m resulted in minor injury, serious injury, or fatality, compared with **30.96%** of crashes beyond 500 m — a difference of only **0.07 percentage points**.
- A chi-square test found no statistically significant association between school proximity and whether a crash involved injury or fatality ($^2 = 0.0017$, p = 0.9671).
- The detailed severity distributions were also similar between the two groups. Overall, the analysis provides no evidence of a higher injury-or-fatal crash proportion near government schools during the selected periods.

9 8. OVERALL FINDINGS

This analysis examined South Australian road crashes from 2020–2024 to investigate whether crash patterns differ during school terms and around school locations.

Four main findings emerged:

9.1 8.1. School terms are associated with distinct temporal crash patterns

- On school-term weekdays, crash frequency shows pronounced peaks around the morning and afternoon travel periods. The largest relative difference occurs at **8 AM**, when the average number of crashes per weekday is approximately **2.1 times** the non-term level.
- A second increase occurs around **3 PM**, when school-term crash frequency is approximately **44% higher** than during non-term weekdays.
- These temporal differences align with typical school travel periods, but the available data do not identify the underlying traffic exposure or travel purpose responsible for the pattern.

9.2 8.2. Spatial evidence is stronger in the afternoon

- At **8 AM**, school-term crashes were descriptively more concentrated within 250 m of the nearest eligible government school than non-term crashes. However, the overall distance distribution did not differ significantly between the two groups ($p = 0.0828$).
- At **3 PM**, school-term crashes were more concentrated near government schools across the examined distance thresholds, and the overall distance distribution differed significantly from non-term crashes ($p = 0.0006$).
- This indicates that the morning pattern is primarily a strong temporal signal, while the afternoon pattern shows clearer spatial evidence of increased concentration around government school locations.

9.3 8.3. Some school areas show relatively high observed crash counts

- The hotspot analysis identified several school areas with relatively high numbers of school-term weekday crashes during the 8 AM and 3 PM periods.
- **Prospect North Primary School** recorded the highest observed crash count in the analysis, followed by **Sturt Street Community School** and **Richmond Primary School**.
- However, crash counts should not be interpreted as direct measures of school safety because surrounding traffic volume, road network design, land use, and other factors were not controlled for.

9.4 8.4. Crashes near schools are not more severe

- Approximately **31.03%** of crashes within 500 m of a school resulted in injury or fatality, compared with **30.96%** beyond 500 m.
- The difference was not statistically significant ($^2 = 0.0017$, $p = 0.9671$).
- Therefore, the main road-safety pattern observed in relation to school-term timing and school proximity concerns crash frequency and spatial concentration rather than crash severity.

9.5 8.5. Overall Interpretation

Taken together, the analysis suggests that school-term crash patterns are characterised more by when and where crashes occur than by increased crash severity. The strongest temporal difference appears at 8 AM, while the clearest spatial concentration near government schools appears at 3 PM.

10 9. RECOMMENDATIONS

Based on the findings, the following actions could support further road-safety investigation and planning:

10.1 9.1. Prioritise morning and afternoon school-travel periods for further road-safety assessment

- The strongest temporal difference was observed at 8 AM, while the clearest spatial concentration near government schools was observed at 3 PM. These periods should therefore receive particular attention in future traffic and road-safety investigations.

10.2 9.2. Prioritise detailed assessment of identified government school areas

- School locations with relatively high observed crash counts, particularly those with a high proportion of crashes within 250 m, can be used to prioritise site-level investigation of surrounding roads, intersections, crossings, and access points.

10.3 9.3. Focus investigation on crash occurrence and exposure rather than assuming greater crash severity near schools

- The analysis found no meaningful difference in injury/fatal crash proportions between crashes within and beyond 500 m of government schools. This suggests that further investigation should focus primarily on why crashes occur at particular times and locations.

10.4 9.4. Incorporate traffic and road-environment data in future analysis

- Traffic volume, pedestrian activity, vehicle speed, road class, intersection design, school entrances, and school enrolment would help determine whether the observed crash concentrations reflect greater exposure, road design, or other local factors.

11 10. LIMITATIONS

- This analysis identifies temporal and spatial patterns in road crashes near government schools, but several limitations should be considered when interpreting the findings.

11.1 10.1. Traffic exposure

- The dataset does not include traffic volume, pedestrian activity, or the number of school-related journeys. Therefore, the analysis compares crash frequency rather than exposure-adjusted crash risk. Higher crash frequency may partly reflect greater traffic activity rather than a higher underlying risk per road user.

11.2 10.2. Proximity and School Attribution

- Crashes are geographically assigned to the nearest eligible government school for spatial analysis. This does not mean that a crash was caused by, belonged to, or was necessarily associated with that school. A crash may also occur within a broader area containing multiple schools.

- School-level results should therefore be interpreted as geographic patterns around government school locations rather than as measures of individual school safety or school-related crash involvement.

11.3 10.3. School coverage

- The education-site dataset includes South Australian Government schools represented in the source data and does not cover all non-government schools. Spatial findings therefore apply specifically to the government school locations included in this analysis.

11.4 10.4. Distance measurement

- Distance to the nearest school is measured using projected straight-line distance. This does not account for the actual road network, school entrances, pedestrian routes, intersections, or physical barriers, which may affect real-world accessibility and exposure.

11.5 10.5. Observational analysis

- This is an observational analysis based on historical crash, school-location, and school-term data. Although differences are observed between school-term and non-term periods, the analysis does not establish a causal relationship between school activity and road crashes.

12 11. CONCLUSION

- This project analysed South Australian road crash data from 2020–2024 together with government school-location and school-term data to investigate temporal and spatial crash patterns around schools.
- The analysis found a clear school-term pattern in crash frequency. School-term weekdays showed a particularly strong increase around **8 AM**, with average crash frequency approximately **2.1 times** the non-term level, while a smaller but notable increase was observed around **3 PM**.
- Spatial analysis showed a different pattern between the two travel periods. At **8 AM**, school-term crashes were descriptively more common within 250 m of the nearest eligible government school, but the overall distance distribution did not differ significantly from non-term crashes. At **3 PM**, school-term crashes showed clearer spatial concentration near government schools, with a statistically significant difference in the overall distance distribution between school-term and non-term crashes.
- Several government school areas were also identified as priority locations for further site-level investigation. However, crashes occurring near schools were **not found to be more severe** than crashes occurring farther away. The proportion of injury or fatal crashes was approximately 31% in both groups, with no statistically significant difference.
- Overall, the results suggest that the main differences in school-term crash patterns relate to crash frequency during school-travel periods and stronger afternoon spatial concentration near school locations, rather than to increased crash severity.
- These findings can help identify times and locations that may warrant more detailed road-safety investigation. However, further analysis incorporating traffic exposure, pedestrian ac-

tivity, road design, and other environmental characteristics would be required before drawing causal conclusions about the role of school-related travel.